Defect-aware graph transformer for impurity formation energies in two-dimensional materials: Chemical transferability and error analysis

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(Dated: August 13, 2026)

Random-split accuracy does not show how defect-energy models transfer across chemistry or how their errors depend on incorporation geometry. We develop the Defect-Aware Radial Transformer (DART), an impurity-centered graph model that combines defect-local message passing, geometric attention, and a non-extensive gated readout. Evaluated on 10,224 neutral impurity structures from IMP2D, DART achieves similar errors under random and pair-held-out evaluation, but the mean absolute error rises to 0.845 and 0.891 eV for unseen impurities and hosts, respectively. The host–impurity asymmetry is associated with structural novelty after accounting for training support. Interstitials are more than twice as difficult as adsorbates: their errors are associated with roomy, weakly constrained environments, whereas adsorbate errors track chemical mismatch, particularly signed valence differences. A readout comparison shows that atomwise addition is poorly matched to the non-extensive target, while defect-centered pooling avoids size-dependent drift. For retrospective reranking of existing DFT-relaxed IMP2D candidates, pair-held-out predictions identify the lower-energy incorporation class with 91.4% accuracy. Atom-relabeling tests further show that an order-sensitive representation can shift individual predictions by up to 8.8 eV while leaving pooled error nearly unchanged. Together, these results connect chemical transfer, local defect environments, and representation correctness in machine-learned defect energetics.

I. INTRODUCTION

Point defects and incorporated impurity atoms reshape carrier densities, localized electronic states, magnetism, and catalytic response in two-dimensional (2D) materials. Their formation energies rank relaxed defective structures relative to a pristine host and an atomic reservoir, and therefore couple the local defect environment to the surrounding chemical reference state [1]. This coupling is especially varied in 2D hosts: defect structures and energetics differ from their bulk counterparts [2], while exposed surfaces and internal voids support distinct adsorbate and interstitial incorporation geometries [3]. The IMP2D database provides a systematic survey of these neutral impurities across host chemistries, impurity elements, incorporation classes, and candidate sites [3]. It thereby enables a broader question than pooled energy prediction: how does model error evolve when the host, impurity, or incorporation geometry departs from the training distribution?

Structure-based machine learning now provides several routes to defect energetics. Graph models operate directly on periodic atomic structures [4, 5] and have been adapted to bulk defects through defect-specific networks, semiconductor models, and persistent-homology features [6–8]. A recent framework extended structural prediction to multiple charge states by aligning reference energies and treating perturbed host states consistently [9]. For 2D systems, sparse defect representations on 2DMD and tree or descriptor models on IMP2D have demonstrated complementary data-efficient strategies [10–12]. Universal machine-learning interatomic potentials offer another route: large-scale calculations of unrelaxed vacancy energies across 86,259 Materials Project entries can be followed by relaxation of selected candidates [13]. These advances establish predictive accuracy, charge-state treatment, and computational throughput, but leave the structure of chemical transfer error largely unresolved.

The central challenge is to distinguish interpolation from transfer across chemistry. A host–impurity pair absent from training may still recombine two familiar constituents, whereas a held-out host or impurity removes an entire source of target supervision. Random structure splits mix these regimes and can also place numerically equivalent structures across the train–test boundary [14]; grouped evaluation is required to separate the resulting levels of chemical novelty [15, 16]. A single pooled error further averages over physically distinct adsorbate and interstitial environments. Finally, a periodic representation must assign identical predictions to structures that differ only by atom labeling. These requirements connect transfer evaluation to both local defect physics and representation correctness.

Impurity formation energy combines a localized structural perturbation with the broader chemistry and geometry of the host, while remaining a non-extensive target. This observation motivates the Defect-Aware Radial Transformer (DART), an impurity-centered graph model that combines defect-local message passing, radial-bias self-attention, and a gated graph readout [Fig. 1]. The local representation is strictly invariant to atom relabeling and uses exact minimum-image distances; the readout fuses defect-sensitive and graph-level information without imposing atomwise extensive scaling. We evaluate the resulting model under random, host–impurity-pair, predefined chemistry-block, impurity-held-out, and host-held-out protocols, followed by class-resolved error analysis.

Across 10,224 retained IMP2D structures, recombining familiar constituents remains close to random interpolation, whereas withholding a complete impurity or host roughly doubles the mean absolute error. The larger host-transfer error is associated with structural novelty after accounting for training support. Incorporation geometry defines a second axis: interstitials are more than twice as difficult as adsorbates, with their errors associated with roomy, weakly constrained environments while adsorbate errors track chemical mismatch. Readout comparisons show that atomwise addition is poorly matched to the non-extensive target. For existing DFT-relaxed IMP2D candidates, pair-held-out predictions identify the lower-energy incorporation class with 91.4% accuracy. Atom-relabeling tests further reveal that an order-sensitive representation can shift individual predictions by up to 8.8 eV while leaving pooled error nearly unchanged. Together, these results connect transferability in impurity energetics to chemical novelty, local defect environments, target scaling, and representation correctness.

II. METHODS

A. Dataset, target provenance, and canonical modeling set

We use neutral impurity structures from the Impurities in Two-Dimensional Materials Database (IMP2D) [3]. The source ASE database contains 17,364 rows. Replaying the data-preparation filter in database order removes 4551 unconverged calculations, 1828 rows with a missing or nonfinite stored formation energy, and 344 rows with $|E_\mathrm{f}| > 20$ eV, leaving 10,641 structures. These structures span 44 hosts, 65 impurity elements, and two incorporation classes (adsorbate and interstitial).

The two incorporation geometries are illustrated schematically in Fig. 2.

For rows with complete source fields, the target was independently recomputed as

$$E_\mathrm{f} = E_\mathrm{def} - E_\mathrm{host} - \mu_\mathrm{imp}. \tag{1}$$

Here E_def and E_host are the total DFT energies of the relaxed impurity-containing and pristine host supercells, respectively. The IMP2D convention takes μ_imp as the per-atom energy of the impurity element in its relaxed crystalline standard state, evaluated with the source workflow [3]. For site or incorporation-class comparisons within one host–impurity pair, E_host and μ_imp are common offsets and cancel. The released relaxed impurity geometry and E_def are products of the same completed IMP2D DFT workflow. Consequently, supplying that geometry does not avoid the relaxation that produced the reference energy. DART receives the relaxed impurity-containing structure rather than the three terms of Eq. (1); it learns their difference implicitly for retrospective energy estimation and reranking, conditional on the available geometry and the IMP2D reference convention. The recomputed and stored values agree to a maximum absolute numerical difference of 2.3×10^{-13} eV. Four otherwise filtered rows contain a zero sentinel in the stored defective-structure energy and therefore cannot be independently reconstructed from the source components; these rows are excluded from every modeling split.

The released relaxed structures do not retain a persistent identity field for the atom initially inserted by the structure generator. In 349 rows, the impurity element is also a constituent of the host, so multiple identical nuclei match the impurity label. Of these cases, 316 are interstitial and 33 are adsorbate structures. Assigning one of them by array order would break permutation invariance and would make the defect-centered features depend on a nonphysical convention. We therefore require exactly one atom to match the impurity element and exclude all 349 non-unique rows from the protocol.

We also prevent detected numerical equivalents from crossing split boundaries. Duplicate candidates are identified by the union of two fingerprints: (i) atomic numbers, cell, and Cartesian coordinates rounded to 10^{-6}, and (ii) a permutation- and translation-invariant spectrum of element-labeled periodic pair distances rounded to 10^{-4} Å. The latter detects numerically equivalent structures even when atom order or coordinate origin differs. Among the remaining eligible structures, connected duplicate groups contain 64 redundant rows; the lowest-index auditable representative is retained. The resulting modeling set contains 10,224 structures. The complete audit, excluded sample indices, database identifiers, and split files are distributed with the code.

B. Fixed evaluation partitions

All partitions were generated once from the curated set and stored as index lists. Five paired random 80/10/10 partitions (assignment seeds 42–46) are used for the 2^3 architecture experiment. A separate fivefold random cross-validation partition provides one out-of-fold prediction for every retained sample. Chemical transfer is evaluated using balanced fivefold group cross-validation in which complete host identities, impurity identities, or host–impurity pairs are held together. In each grouped fold, one group fold is used for testing and the next group fold for validation. We further note that pair grouping does not mathematically require each marginal host and impurity to remain in training. A split audit found one singleton-host test structure for which this condition fails; the other 10,223 pair-held-out test structures retain both constituent identities in their corresponding training fold. We further use a predefined matrix-completion holdout: group-VI dichalcogenide hosts combined with $3d$ impurities form the test block, while group-IV/V hosts combined with $4d$ impurities form the validation block. After the common audit exclusions, this partition contains 9,487 training, 372 validation, and 365 test structures.

Uncertainty quantification uses a separate random par-

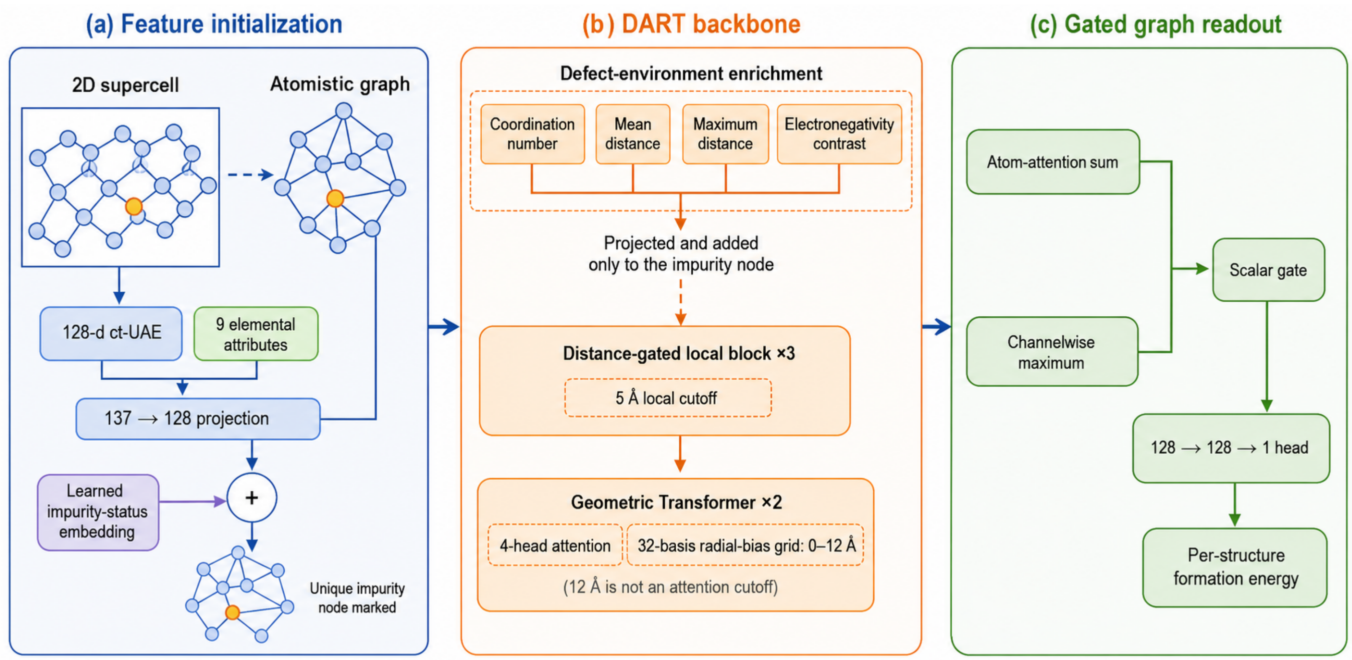

FIG. 1. Architecture of the Defect-Aware Radial Transformer (DART). (a) A fixed 128-dimensional ct-UAE vector and nine normalized elemental attributes are projected from 137 to 128 channels, after which a learned impurity-status embedding marks the unique impurity node. (b) Four projected defect-environment quantities are added only at that node before three 5-Å distance-gated local blocks and two four-head geometric Transformer blocks. The Transformer radial bias uses 32 basis functions spanning 0–12 Å; 12 Å is the end of the basis grid rather than an attention cutoff. (c) A graph-dependent scalar gate fuses an atom-attention weighted sum and a channelwise maximum before a $128 \to 128 \to 1$ head predicts one formation energy per structure. The schematic was edited with OpenAI GPT Image 2; the authors specified and verified its scientific content.

tition with 75% for training, 10% for checkpoint selection, 5% for calibration, and 10% for internal testing (assignment seed 62). Calibration and test indices are excluded from ensemble training and checkpoint selection, and test labels are not used for calibration. All split files cover the full 10,641-row container through mutually disjoint train, validation, test, optional calibration, and exclusion sets.

The sequential audit, retained chemical coverage, target distribution, and fixed partition geometry are summarized in the Supplemental Material [17].

C. Defect-aware graph Transformer

Each periodic supercell is represented by an atomistic graph. We refer to the resulting model as the Defect-Aware Radial Transformer (DART). DART uses the defect environment as a local inductive bias while retaining global host context. A uniquely identified impurity node exposes the defect center, short-range message passing represents its coordination environment, periodic radial attention supplies longer-range response, and graph-level readout maps the combined representation to one formation energy. The initial node representation concatenates a fixed 128-dimensional ct-UAE elemental vector, reconstructed verbatim from the published checkpoint, with nine min–max-normalized elemental attributes: group, period, Pauling electronegativity, covalent and van der Waals radii, a group-based valence count, first ionization energy, electron affinity, and atomic mass [18]; the exact reconstruction and feature-table conventions are specified in the Supplemental Material [17]. The concatenated representation is projected to 128 hidden channels, and a learned binary embedding marks the impurity atom. Three local interaction layers aggregate neighbors within 5 Å. Distances are expanded in 32 Gaussian radial basis functions, and three-body messages use a 32-function angular expansion. Two subsequent four-head geometric Transformer blocks apply self-attention to all valid atoms, with a learned radial bias constructed from 32 basis functions centered between 0 and 12 Å [19]. A two-layer multilayer perceptron maps the pooled graph representation to $\hat{E}_{\mathrm{f}}$. Because every retained row passes the identity audit, the binary impurity embedding always marks exactly one compositionally unique atom; no array-order heuristic is used. Figure 1 summarizes this data path, from the 5-Å local block through the radial-biased geometric Transformer to gated graph readout.

The confirmatory architecture experiment varies three binary modules while holding every other training and model setting fixed:

Gated readout (G).: A learned atom-attention sum

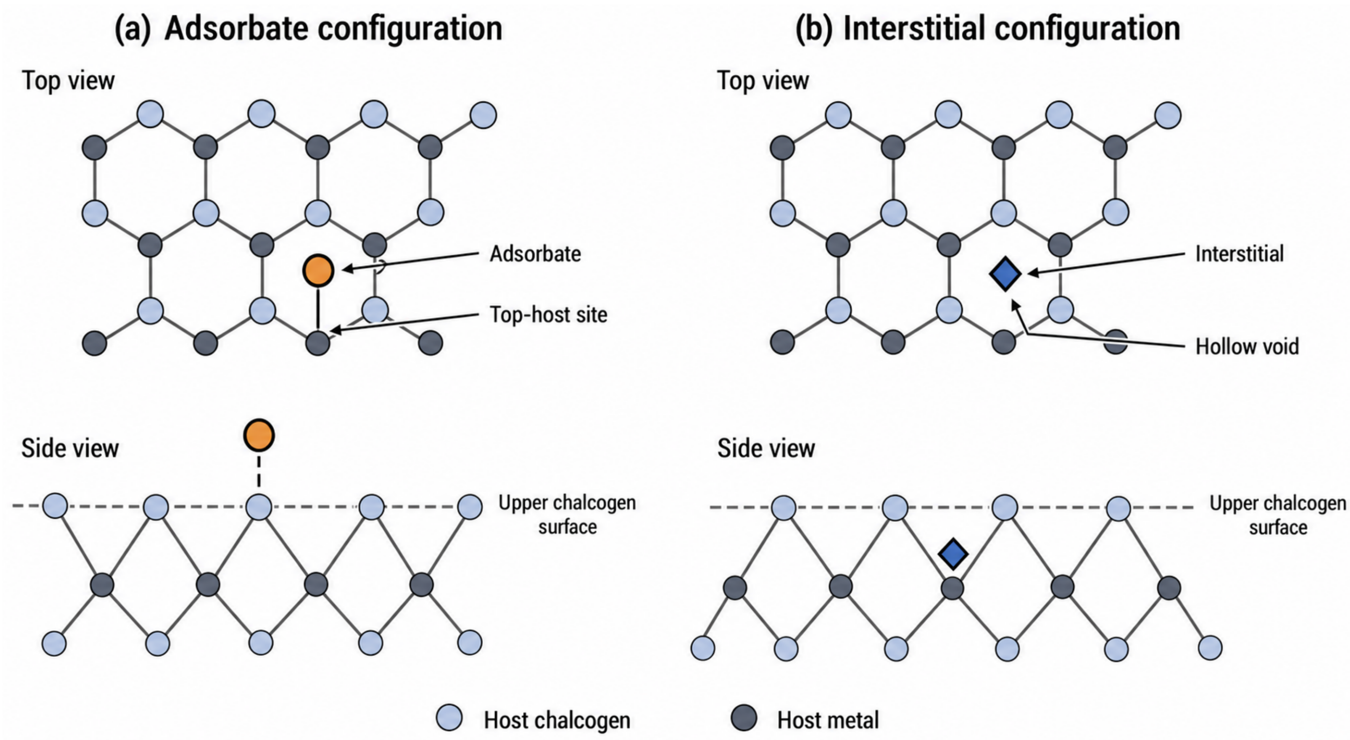

FIG. 2. Structural representation of the two neutral impurity incorporation classes in two-dimensional host materials. (a) Top and side views of an adsorbate above a top-host site. (b) The corresponding views of an interstitial impurity in a hollow void below the upper chalcogen surface. Orange circles and blue diamonds encode the two classes by both color and shape. The panels are alternative single-impurity configurations in a WSe_2-like host, not coexisting dopants, and illustrate representative rather than exhaustive IMP2D sites. The schematic was edited with OpenAI GPT Image 2; the authors specified and verified its scientific content.

and a channelwise maximum are fused by a graph-dependent scalar gate. The off state uses masked mean pooling.

Environment enrichment (E).: Four defect-centered quantities—coordination number, mean and maximum neighbor distance, and the absolute electronegativity contrast to the local neighborhood—are normalized, projected, and added only to the impurity representation. The off state omits this branch.

Pre-normalized, distance-gated local block (P).: The on state normalizes node states before message construction and applies a learned scalar distance gate to each edge before a residual update. The off state uses the original ungated message and post-residual normalization [20]. We treat this as one composite local-block change and do not attribute its effect to normalization or edge gating separately.

For the selected local block, let $\widetilde{\mathbf{h}}_i$ be the layer-normalized state of atom i, $\mathbf{r}(d_{ij})$ its radial basis, and g_{ij} a learned scalar gate. Pair and angle messages are

$$\mathbf{m}_{ij} = \phi_r[\mathbf{r}(d_{ij})] \odot W_v \widetilde{\mathbf{h}}_j \, g_{ij}, \quad g_{ij} = \sigma\{\phi_g[\mathbf{r}(d_{ij})]\}\,, \tag{2}$$

$$\mathbf{t}_{jik} = \phi_\theta[\widetilde{\mathbf{h}}_i \| \mathbf{a}(\theta_{jik})], \tag{3}$$

where $\mathbf{a}$ is the angular basis and $\|$ denotes concatenation. The graph builder forms the complete ordered-angle multiset at each center and, when its size $n = d(d-1)$ exceeds 32, retains the midpoint order statistics $\lfloor (2q+1)n/64 \rfloor$, $q = 0, \ldots, 31$, of the sorted angles; the selection depends only on angle values, never on stored atom order, so the retained set $\mathcal{T}_i$ is invariant to atom-index permutations. The residual update is

$$\mathbf{h}'_i = \mathbf{h}_i + \phi_n\left(\sum_j \mathbf{m}_{ij} + \sum_{(j,k)\in\mathcal{T}_i} \mathbf{t}_{jik}\right). \tag{4}$$

For attention head a, the subsequent global block uses

$$A_{ij}^{(a)} = \mathrm{softmax}_j\left[\frac{\mathbf{q}_i^{(a)} \cdot \mathbf{k}_j^{(a)}}{\sqrt{d_h}} + b_a\{\mathbf{r}_g(D_{ij})\}\right], \tag{5}$$

where D_{ij} is the exact minimum-image distance under the structure's periodic boundary conditions, computed with a verified shortest-image construction and validated against a brute-force image search on orthogonal, hexagonal, and adversarial oblique cells. The 32 global radial basis centers span 0–12 Å; beyond the outer center their activations decay toward zero, while attention still connects every valid atom. The gated readout is

$$\mathbf{h}_{\mathrm{att}} = \sum_i \alpha_i \mathbf{h}_i, \tag{6}$$

$$\mathbf{h}_{\max} = \max_i \mathbf{h}_i, \tag{7}$$

$$s = \sigma\{\phi_s(\mathbf{h}_{\mathrm{att}} \| \mathbf{h}_{\max})\}, \tag{8}$$

$$\mathbf{h}_G = \mathrm{LN}[s\mathbf{h}_{\mathrm{att}} + (1-s)\mathbf{h}_{\max}]\,. \tag{9}$$

Elemental scalars, local distances and angles, and minimum-image global distances make the scalar prediction invariant to rigid translation and rotation of a periodic structure, and every aggregation — pair messages, the order-free triplet selection, attention, and graph pooling — is symmetric under atom-index permutations. Both properties are enforced by unit and property tests and were re-verified in the independent acceptance audit of the released implementation.

The full 2^3 factorial contains all eight G/E/P combinations on each of the five paired random partitions. Within a partition, all variants share the same initialization seed and training batches. Main effects and interactions are orthogonal factorial contrasts. The architecture used in subsequent transfer and uncertainty experiments is the variant with the lowest mean validation MAE; exact ties are resolved by fewer enabled modules and then by binary variant code. Test metrics are not part of the ordering rule.

D. Training protocol

Networks are optimized for 150 epochs with AdamW, a peak learning rate of 5×10^{-4} reached after a ten-epoch linear warmup from 5×10^{-5}, weight decay 10^{-4}, and cosine decay to 10^{-6}. The objective is mean absolute error (MAE), batch size is 64 for DART, and gradient norms are clipped at 5. Targets are standardized using training-partition statistics for optimization and transformed back to eV for all reported errors. The fixed DART regularization recipe is limited to 0.1 dropout and additive Gaussian target noise with standard deviation 0.03 eV; no coordinate or strain augmentation is applied. A host-balanced sampler draws 200 structures per host per epoch; its sequence and the independent target-noise stream are deterministically indexed by model seed and epoch so paired variants see the same batches and noise realizations. Every factorial variant uses the same two input assets. The ct-UAE table described above is a non-trainable buffer. Separately, we pretrained the nine-feature input projection and three ungated, post-normalized local interaction layers for 30 epochs. We first drew 20,000 records uniformly without replacement from JARVIS-DFT `dft_3d` with seed 42; conversion and graph-validity filters retained 19,902 pristine three-dimensional crystals [21]. Pretraining used formation energy per atom as the source target, an 80/10/10 random partition, and seed 42. No IMP2D target is used in this stage. An explicit chemistry-overlap audit found that 26 of the 44 IMP2D host labels (25 of 42 distinct reduced host formulas) occur among 69 JARVIS source records, and that all 65 IMP2D impurity elements occur somewhere in the source-task structures. Formula overlap does not establish structural identity, but it means that the grouped partitions withhold IMP2D formation-energy supervision rather than all prior source-task exposure to the same reduced host formula or impurity element. Accordingly, "host held out" and "impurity held out" denote absence from IMP2D target supervision, not absence from the JARVIS source-task chemistry.

The checkpoint initializes the nine-feature slice of the input projection, its bias, and all compatible parameters of those three local-interaction layers. The ct-UAE slice and the P-specific edge gates retain their seeded initialization. All trainable parameters are then optimized on the assigned IMP2D partition. We select the epoch with minimum validation MAE and evaluate it once on the corresponding test partition. Before aggregation, partition membership is checked and all regression metrics are recomputed from the stored targets and predictions.

E. Baselines

Two baseline classes use exactly the same partitions. First, an 80-dimensional deterministic descriptor uses atomic number, mass, covalent radius, and Pauling electronegativity. Mean, standard deviation, minimum, maximum, and median over all atoms and host atoms contribute 40 entries; impurity values and impurity-minus-host-mean contrasts contribute eight. Eighteen entries encode atom count, elemental diversity, cell lengths, area, volume, mass density, the first six impurity-neighbor distances, two coordination counts, and local-distance mean and standard deviation. Twelve site indicators and two incorporation-class indicators complete the vector. Missing electronegativities use the fixed value 2.0 and fewer than six neighbors are padded at 20 Å. We fit a training-target mean predictor, ridge regression, random forests, histogram gradient boosting, and LightGBM [22]. These model families also permit direct comparison with earlier IMP2D descriptor studies [11, 12]. Hyperparameters within each family are selected by validation MAE only. The learned descriptor family is then chosen independently inside every split by that split's validation MAE; pooled scores concatenate the corresponding fold-wise selected test predictions. Thus no held-out fold's test labels enter the family used to predict that fold. The training-target mean predictor is reported separately and is not a candidate in family selection.

Second, the predefined additive-readout periodic SchNet recipe [5] uses 128 hidden channels, six interaction blocks, 50 Gaussian functions, and a 5 Å cutoff. Its graph prediction sums learned atomwise scalar contributions, following the additive energy readout of the original SchNet formulation. It uses batch size 32 but otherwise follows the same target normalization, 150-epoch AdamW schedule, host balancing, gradient clipping, and target-noise protocol as DART; it uses a conventional learned atomic-number embedding and no DART pretraining. Random and host–impurity-pair fivefold evaluation uses one fixed model seed per fold (242 for DART and 342 for SchNet). Host-held-out, impurity-held-out, and chemistry-block evaluation uses three independently initialized seeds (242–244 for DART and 342–344 for SchNet). Predictions are averaged within each split before pooled out-of-fold metrics are calculated. Consequently, DART–SchNet contrasts compare the complete learning recipes, including their different input representations, initialization, and graph readouts; they do not isolate a pure backbone-architecture effect. The selected DART has 820,558 trainable parameters, compared with 455,937 for SchNet.

To assess the unusually large host-held-out SchNet error, we repeated that evaluation after replacing only atomwise addition with mean pooling. All other model, optimization, split, and seed settings were unchanged. Predictions were averaged over the three seeds within each fold, and the mean-minus-add difference in absolute error was assigned a 95% percentile interval from 20,000 host-cluster bootstrap samples. This post hoc sensitivity is reported separately from the main benchmark.

F. Internal uncertainty calibration and retrospective reranking

Five independently initialized members of the validation-selected DART architecture are trained on the dedicated uncertainty split, following the deep-ensemble construction and its atomistic use as a model-credibility signal [23, 24]. Ensemble mean and sample standard deviation provide the point prediction and a raw model-disagreement proxy. A fixed-seed random permutation (seed 6201) assigns the dedicated calibration rows to two subsets without using target labels. One subset fits a nonnegative scale and floor for

$$\sigma_i = \sqrt{(as_i)^2 + b^2} \tag{10}$$

by Gaussian negative log likelihood; the other subset estimates finite-sample split-conformal quantiles of $|E_{\mathrm{f},i} - \hat{E}_{\mathrm{f},i}|/\sigma_i$ [25, 26]. Under exchangeability within this split, the resulting intervals have finite-sample marginal coverage at each evaluated level; the guarantee is not conditional on host or impurity identity. Because architecture development used other partitions of the same corpus, we interpret the reported coverage as internal calibration rather than as an untouched outer-holdout estimate. Test labels are not used in calibration.

For retrospective reranking, the five host–impurity-pair folds yield one out-of-fold prediction per retained structure. Within each host–impurity pair we compare the minimum predicted and reference energies of adsorbate and interstitial configurations, identify the predicted lowest-energy site, and report preference accuracy and reranking regret. We define the incorporation-class margin as

$$\Delta E_{\mathrm{cls}} = E_{\mathrm{min}}^{\mathrm{int}} - E_{\mathrm{min}}^{\mathrm{ads}}, \tag{11}$$

so positive values favor adsorption. Reranking regret is the excess reference energy of the configuration selected by the prediction over the reference minimum in the eligible candidate set. Exact-site metrics require at least two candidate sites, whereas top-two accuracy requires at least three; reference minima within 10^{-8} eV are treated as tied, and tied adsorbate–interstitial pairs are excluded from binary preference accuracy.

Accuracy is contextualized by two non-trained descriptive decision references defined over the eligible evaluation decisions. Incorporation class uses the empirical majority class among eligible IMP2D labels. Exact-site accuracy uses the analytic expectation from choosing uniformly among the observed candidates, and top-two accuracy uses the corresponding uniformly sampled size-two subset; tied reference minima count as successful choices. Model-minus-reference intervals use the same host–impurity-pair cluster bootstrap as the model estimates. These are predictive associations within IMP2D, not causal mechanism assignments or external DFT validation.

III. EVALUATION PROTOCOL

A. Metrics and statistical analysis

We report MAE, root-mean-square error (RMSE), signed bias, Spearman rank correlation, and R^2. Host- and impurity-macro MAE give each chemical group equal weight, independent of its number of structures. Cross-validation predictions are concatenated only when test folds are disjoint; model seeds are averaged within a fold before concatenation. Low-energy behavior is assessed by MAE for $E_{\mathrm{f}} \leq 0$, MAE within the true lowest-energy decile of each pooled test regime, and recall of that decile by the predicted lowest-energy decile. These strata are evaluation summaries and are not used for architecture or hyperparameter selection.

For the architecture factorial, each effect is evaluated as the difference between the average metric at positive and negative levels of the associated orthogonal contrast within each paired repeat. Ninety-five-percent intervals are obtained by nonparametric bootstrap over the five repeat-level contrasts. Because only five paired repeats were used, the intervals are accompanied by repeat-level sign consistency and interpreted as descriptive uncertainty summaries rather than large-sample hypothesis tests. For model comparisons on aligned out-of-fold predictions, we use 10,000 paired bootstrap resamples. The resampling unit matches the held-out unit: individual structures for random cross-validation, complete hosts for host-held-out evaluation, complete impurities for impurity-held-out evaluation, and host–impurity pairs for pair-held-out and chemistry-block evaluation. Within each resampled chemical unit all associated structures remain together. Reranking intervals use 20,000 host–impurity-pair cluster-bootstrap resamples. Adsorbate and interstitial site decisions from the same pair remain together, while point estimates average all eligible decision sets. An interval excluding zero is treated as evidence for a directional difference; otherwise the result is reported as inconclusive rather than equivalent.

Uncertainty evaluation includes empirical interval coverage and mean width at 50%, 80%, 90%, and 95% nominal coverage, Gaussian negative log likelihood, the continuous ranked probability score, Spearman correlation between uncertainty and absolute error, and risk–coverage curves. The area under the risk–coverage curve (AURC) is accompanied by its excess over oracle ordering [27]. All calibration choices use only the dedicated calibration partition.

All energy estimates and reranking metrics are retrospective against the retained IMP2D targets and conditioned on geometries produced by the same DFT workflow as the reference energies. The study does not evaluate an independently generated, target-matched first-principles test set or claim to replace candidate generation and relaxation.

IV. RESULTS

A. Chemical transfer is hierarchical, and the hierarchy tracks structural novelty

Withholding chemistry degrades prediction in a strict order. With the repaired graph representation, mean absolute error is 0.408 eV under random cross-validation, rises modestly to 0.442 eV when complete host–impurity *pairs* are withheld and to 0.483 eV on the predefined host-by-impurity chemistry block, but jumps to 0.845 eV when complete impurities are withheld and to 0.891 eV when complete hosts are withheld. Recombining constituents that were each seen in training is therefore only mildly harder than interpolation, whereas encountering a new constituent roughly doubles the error, and a new host is the hardest axis of all. Figure 3 first shows the class-resolved pair-held-out prediction cloud and then places the five regimes and the two aligned comparators — the additive-readout SchNet under the identical protocol and the descriptor model selected by validation MAE within each split — on a common scale; numerical values are tabulated in the Supplemental Material [17]. The defect-centered model leads every regime; the SchNet gap widens sharply on the constituent-transfer axes, while the descriptor gap is roughly scale invariant at 1.3–1.6×, so the constituent-transfer difficulty is shared by all three model families rather than an artifact of one architecture.

The host–impurity ordering itself is small but real: the sample-matched host-minus-impurity absolute-error gap has a pair-clustered 95% interval of [0.010, 0.085] eV ($p = 0.011$). To ask *why* host holdout is harder, we constructed fold-local novelty scores — structural novelty of the held-out host from composition, cell, thickness, and coordination descriptors, and chemical novelty of the held-out impurity from radius, electronegativity, valence, and period/group — together with a training-support covariate that captures the mechanical asymmetry between the two protocols. The error gap tracks the host's structural novelty: the novelty-gap slope is 0.0010 eV per robust unit (95% interval [0.0007, 0.0013]) and is unchanged when the training-support covariate is included ($p = 0.001$), so the association is not an artifact of the mechanically smaller per-row training support of the host protocol. The preregistered verdict is *consistent with structural-motif shift*: a held-out host confronts the model with unseen structural motifs, whereas a held-out impurity is still surrounded by familiar hosts.

B. Interstitial error is a local-crowding effect; adsorbate error is a chemical-mismatch effect

Under pair-held-out prediction the two incorporation classes fail for different, measurable reasons. Fitting the out-of-fold error against frozen local geometry and chemistry descriptors — with host, impurity, and fold absorbed as crossed effects, and every contrast preregistered before any outcome was viewed — yields seven associations that pass the main-text gate (adjusted decile contrast of at least 0.10 eV or 20%, clustered interval excluding zero, direction stable in at least four of five folds, and surviving the exclusion of pathological out-of-plane reconstructions).

For *adsorbates*, error tracks chemistry, led by signed valence mismatch: moving from the tenth to the ninetieth percentile raises the adjusted error by 0.317 eV, with local geometry contributing through the smoothed coordination measure (the raw five-Ångström count shown as the frozen profile feature in Fig. 4 carries no gated adsorbate effect). For *interstitials*, the five gated effects share one consistent — and initially counterintuitive — direction: error *rises* with post-relaxation clearance (0.091 eV) and *falls* with neighbor-distance dispersion (-0.103 eV), with the largest neighbor distance, covalent-radius mismatch, and valence mismatch following the same pattern, contrast by contrast in Fig. 4(b). Interstitial error therefore concentrates not in tight fits but in *roomy, uniformly spaced, loosely constrained* cages with mismatched valence — environments where the impurity is weakly held and snug packing constraints do not pin the energy. Fig-

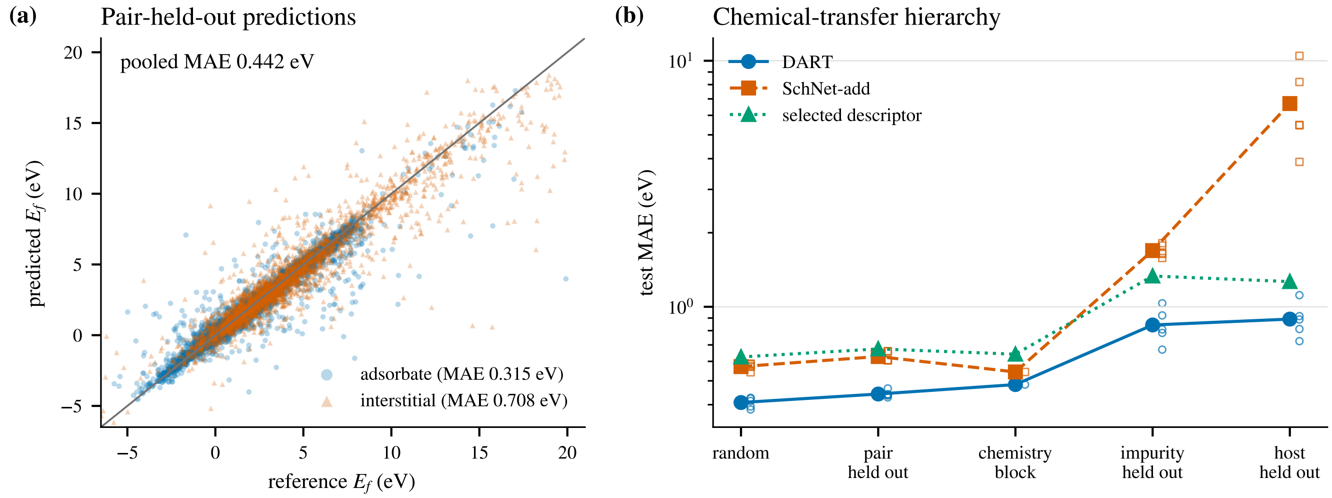

FIG. 3. Prediction accuracy and chemical-transfer hierarchy. (a) Repaired pair-held-out out-of-fold predictions for all 10,224 retained structures, resolved by incorporation class; the diagonal denotes exact agreement. (b) Test MAE across the five transfer regimes for DART, the additive-readout SchNet baseline, and the validation-selected descriptor model on a logarithmic scale. Small open markers show individual folds where five folds exist.

ure 4 reports the gated physical error map: class-resolved adjusted error profiles over coordination and clearance, and the chemistry-mismatch decile contrasts. The out-of-plane expansion-factor diagnostic behind the XF ≤ 2 gate and the family-level macro errors are reported in the Supplemental Material [17].

The asymmetry is large in absolute terms: under pair-held-out prediction, interstitial structures have MAE 0.708 eV against 0.315 eV for adsorbates — a factor of 2.2. The largest surviving interstitial errors occur for extreme-formation-energy cases such as Cr@$ZrSe_2$ and Li@WTe_2; the Supplemental Material [17] renders them alongside matched adsorbate controls as illustrations rather than gated evidence.

To quantify how much of the interstitial penalty the measured environment explains, we compare the incorporation-class coefficient before and after adding the full geometry/chemistry block to a controls-only model. The class gap itself is firmly established (controls-only class coefficient 0.53 in log-error units, interval [0.48, 0.58]). Adding the full block attenuates it to 0.38 [0.22, 0.54] — a 28% point-estimate reduction with the same sign in all five folds, but the attenuation interval [−0.01, 0.58] does not exclude zero, so under the frozen ladder the interstitial penalty is *not explained* by the available descriptors: local crowding accounts for at most a modest fraction, and the remainder is an open question for descriptors beyond this frozen set.

Host-family decomposition places the largest interstitial errors in the single-identity oxide host and the carbide/MXene-like family, with halide/halochalcogenide hosts next; among impurity series, f-block and main-group interstitials are hardest. The retained host set contains no nitride, so no nitride panel is reported. Per-element errors and counts, family macro errors with identity counts, the out-of-plane expansion diagnostic, and rendered extreme-case and matched-control structures selected by the frozen case rule are given in the Supplemental Material [17].

C. Formation energy is not atomwise additive, and the readout must respect that

The impurity formation energy is a difference of extensive totals and is therefore a *local* target: its magnitude does not grow with supercell size. A sum readout imposes extensive scaling on the prediction, and the data show the consequence directly. Under the same training protocol and host-held-out splits, the additive-readout SchNet baseline reaches a pooled MAE of 6.703 eV; replacing only its readout by a mean reduces this to 2.552 eV, still well above the defect-centered model. The preregistered extensivity diagnostic makes the mechanism explicit: the additive arm's signed residual grows by 0.038 eV per atom at the point estimate — fifty times the repaired defect-centered model's 0.0007 eV per atom, with the mean readout at −0.011 eV per atom — but the host-clustered 95% interval spans zero ([−0.213, 0.272] eV per atom; three of five folds share the sign), so under the frozen gate the extensivity explanation is *not established* and the readout conclusion remains an empirical complete-recipe result. These comparisons are complete-recipe contrasts under one protocol; the readout sensitivity is a diagnostic, not a universal backbone ranking.

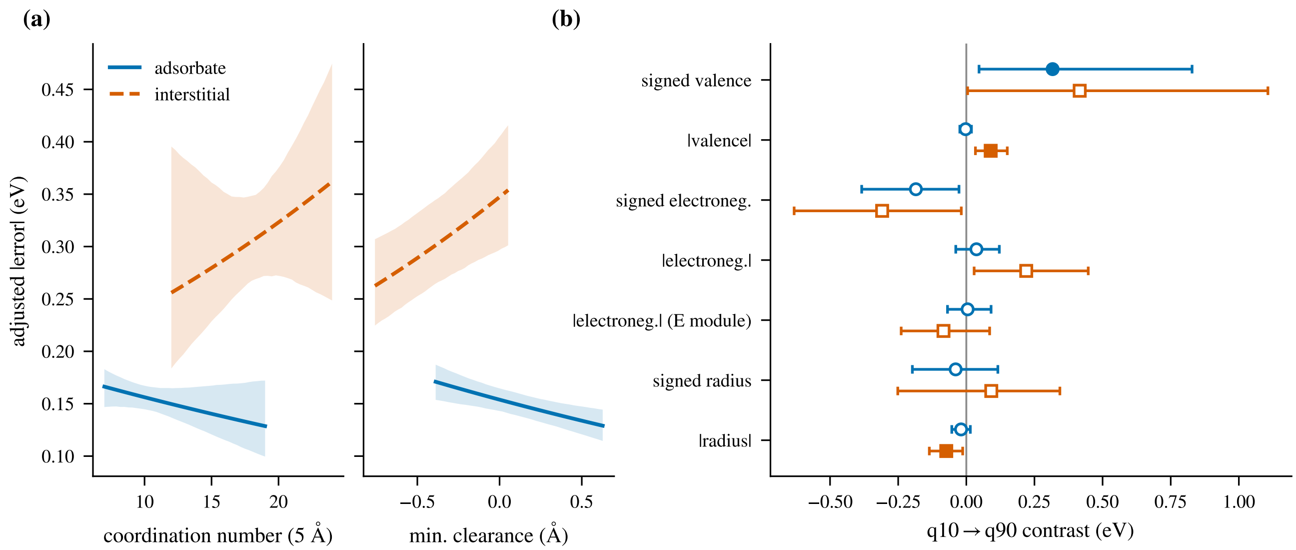

FIG. 4. Physical error map on repaired pair-held-out predictions. (a) Adjusted absolute-error profiles versus smooth coordination and post-relaxation clearance, by incorporation class, with pointwise intervals. (b) Chemistry-mismatch decile contrasts with pair-clustered intervals; filled markers pass the preregistered main-text gate, open markers do not.

D. Retrospective out-of-fold reranking within IMP2D

The host–impurity-pair folds provide one out-of-fold prediction for every retained structure. Both the input geometries and reference energies come from the completed IMP2D DFT workflow, so the following comparisons are retrospective rather than a substitute for site generation or relaxation. Across pairs containing both incorporation classes, the repaired model identifies the lower-energy class among the retained, DFT-relaxed IMP2D candidates with 91.4% accuracy against an empirical majority-class reference of 66.2% — a gain of 25.1 percentage points [22.7, 27.6] — and the predicted and reference class margins have Spearman correlation 0.957. With the incorporation class fixed, exact-site and top-two accuracies are 73.6% and 91.5% against references of 37.1% and 62.1%, with mean reranking regret 0.099 eV. Complete decision metrics with pair-clustered intervals are reported in the Supplemental Material [17]. Figure 5 shows the class-margin agreement, the regret distributions, and the clustered decision-accuracy intervals; class flips concentrate where the reference margin itself is small.

E. Supporting analyses

The architecture factorial that selected the frozen recipe, the internal ensemble uncertainty analysis, and all cutoff and threshold sensitivities are reported in the Supplemental Material [17]; they use the frozen recipe unchanged and none of their numbers supports a main-text claim.

V. DISCUSSION

A. Physical reading of the error structure

The transfer hierarchy has a structural reading. Recombining familiar constituent representations is nearly free; the expensive step is a new constituent, and the preregistered novelty analysis attributes the host–impurity asymmetry: the sample-matched error gap follows the structural novelty of the held-out host — its composition, cell, thickness, and coordination environment relative to the training hosts — and survives controlling for the mechanical training-support asymmetry between the two protocols. A new host changes the structural motif the defect embeds into, whereas a new impurity largely re-poses a chemistry seen in other combinations. Development-era case studies of five named held-out hosts (Supplemental Material [17]) trace the same gradient concretely: hosts with close structural relatives in training transfer at no measurable cost, while the one host with no structural relative in the database sits at the failure extreme.

These grouped tests do not represent element- or formula-disjoint pretraining. The source task contains reduced-formula matches for 26 IMP2D host labels and every IMP2D impurity element, so host- and impurity-held-out mean absence from formation-energy supervision, not absence from all earlier representation learning. All conclusions are internal to the retained 44 hosts and 65 impurities — a set that contains no nitride host — and to the IMP2D reference conventions.

The class asymmetry is likewise not one phenomenon. A snugly caged impurity sits in a steep, sterically pinned energy well whose depth follows the local geometry the

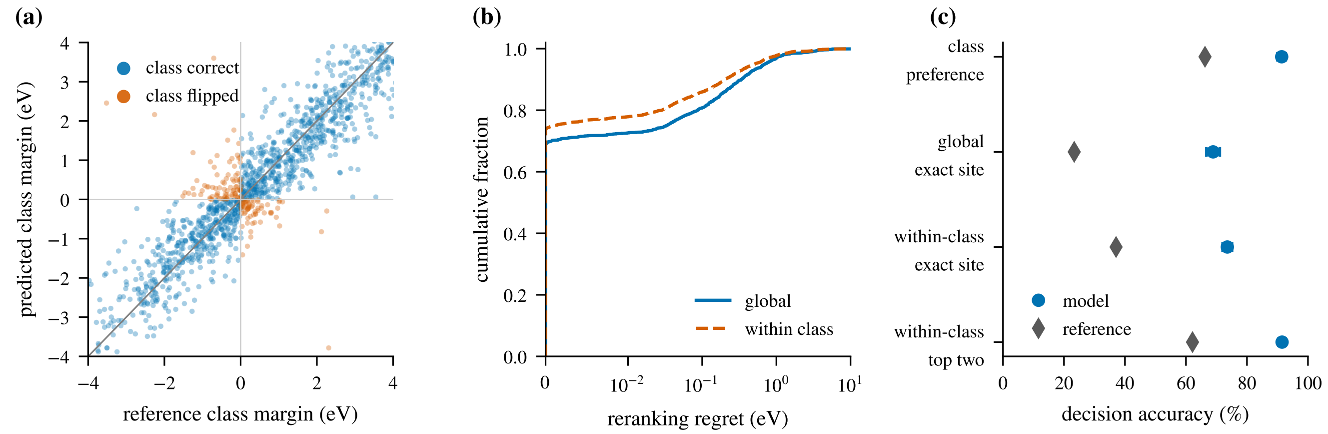

FIG. 5. Retrospective reranking of available DFT-relaxed IMP2D geometries from repaired pair-held-out predictions. (a) Predicted versus reference incorporation-class margins; flipped decisions (orange) cluster at small reference margins. (b) Empirical cumulative distributions of global and within-class reranking regret. (c) Host–impurity-pair cluster-bootstrap intervals for the four decision accuracies against their class-prior or uniform references (diamonds).

model sees directly; a loosely held impurity in a large, uniform cage samples a flatter landscape with near-degenerate placements, where the energy depends on subtler electronic contributions — which is why interstitial error concentrates in roomy, loosely constrained cages while adsorbate error tracks chemical mismatch. These are controlled associations, not causal mechanisms; the sign pattern is what the frozen analysis established. The measured descriptors attenuate the class gap by roughly a quarter at the point estimate, but the attenuation interval includes zero, so the penalty is reported as not yet explained: the remainder must live in physics the frozen descriptors do not span, such as the anisotropy of the relaxation pattern, which the released database does not record. The impurity-identity audit also makes the retained population selective (316 interstitial and 33 adsorbate rows removed), so the asymmetry applies to the retained population rather than to every released IMP2D row.

The readout result follows the same defect-local logic. The formation energy of Eq. (1) is a defect excess energy — a difference of extensive totals that does not grow with host atom count — so atomwise additive pooling imposes the wrong scaling class, and pooling choice is known to matter for intensive and localized targets [28]. The extensivity diagnostic supports this mechanism at the residual level, with a positive additive residual-size slope whose host-clustered interval nevertheless spans zero, so the mechanism is suggested rather than established; development-era attribution analyses (Supplemental Material [17]) show the same property from inside the model, with occlusion attributions concentrated at the impurity atom and decaying with distance from it. These are complete-recipe contrasts under one protocol, not a universal backbone ranking.

B. Scope of the reranking claim

For a fixed host–impurity pair, the pristine-host energy and impurity chemical potential in Eq. (1) cancel when sites or incorporation classes are ranked, which is why useful relative decisions coexist with a larger absolute error. The decisions are conditional on the DFT-relaxed candidates released by IMP2D: each geometry and its reference energy are outputs of the same completed workflow, so no relaxation cost is bypassed, and the results concern retrospective energy estimation and reranking — not structure generation, relaxation, thermodynamic populations, kinetics, or synthesizability.

C. Correctness, uncertainty, and remaining scope

Every number in this paper comes from a repaired representation whose permutation invariance and exact minimum-image distances passed an independent acceptance audit; the source-task pretraining was rebuilt under the same corrected construction. The diagnostic of the superseded implementation is a result in its own right [Fig. 6]: sixteen atom-relabeling permutations moved single predictions by up to 8.8 eV and flipped incorporation-class decisions, while the pooled MAE moved by less than 10^{-3} eV. Aggregate benchmarks are therefore nearly blind to representation defects that dominate individual predictions — a caution we believe generalizes beyond this system.

The evidence remains limited to neutral adsorbates and interstitials, the source database's relaxed structures and reference energies, and a model that infers the reference-energy difference from the impurity-containing graph alone. Vacancies, substitutions, charged defects, transition levels, migration barriers, finite-temperature

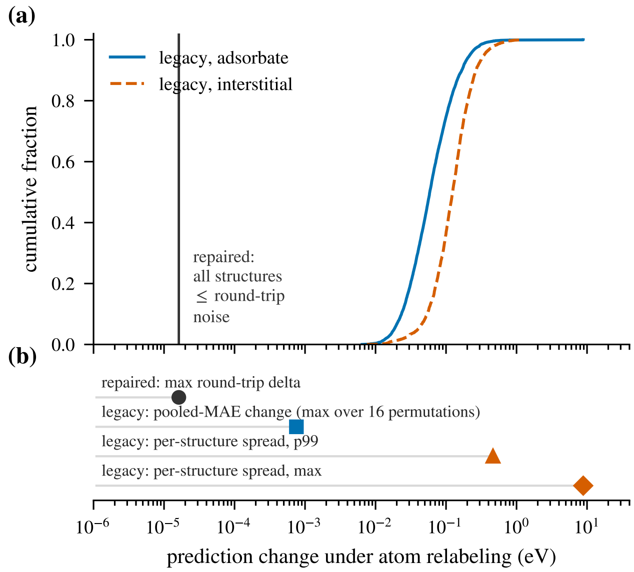

FIG. 6. Atom relabeling as a representation diagnostic. (a) Cumulative distribution of the per-structure prediction spread of the superseded (legacy) implementation across sixteen atom permutations, by incorporation class ($p_{99} = 0.46\,\text{eV}$, maximum $8.8\,\text{eV}$); under the repaired representation every structure reproduces its prediction to within the archived round-trip bound ($\leq 1.7 \times 10^{-5}\,\text{eV}$, vertical line). (b) The same axis at aggregate level: the legacy pooled MAE changes by at most $8 \times 10^{-4}\,\text{eV}$ across permutations — four orders of magnitude below the largest single-structure spread — while individual relabelings flip 9–19 incorporation-class decisions and 76–102 site selections. Aggregate benchmarks are nearly blind to this defect.

free energies, and competing phases are outside scope. Duplicate control removes numerical equivalents detected by the two declared structural fingerprints but cannot prove that every physical redundancy has been found. SchNet and the descriptor families are aligned comparators under the frozen protocol, not an exhaustive survey; a development-era survey (Supplemental Material [17]) additionally evaluated trained equivariant models (MACE, ViSNet), ALIGNN, and the MACE-MP-0 foundation model as a zero-shot label surrogate — the latter essentially uncorrelated with the IMP2D formation-energy convention — and none approached the defect-centered design at matched scale, consistent with reports that universal interatomic potentials degrade on interstitial-like environments far from their bulk training distributions. In absolute terms, the repaired random-split and pair-held-out errors sit in the few-tenths-of-an-eV band reported for specialized defect formation-energy models on bulk and low-dimensional benchmarks [6, 9, 10]; the grouped regimes report the additional cost of chemical novelty that pooled numbers do not expose. Internal ensemble uncertainty (Supplemental Material [17]) supports marginal calibration and risk ranking within the retained population, not host-conditional or chemically shifted coverage.

The single-database scope is itself an evidence-based protocol choice. Development-era experiments (Supplemental Material [17]) found zero-shot cross-database evaluation on JARVIS vacancy data failing at the 2–3 eV level with negative R^2; few-shot fine-tuning lost to training from scratch; and multi-source joint training bought small in-distribution gains at the price of degraded held-out-host transfer. The barrier is one of reference conventions and defect-class mismatch. The most direct next test is therefore a target-matched external set: structures, chemical potentials, convergence settings, and target definition matching the IMP2D convention, generated independently of their reference energies. A development-era pilot (Supplemental Material [17]) has already exercised that test end to end with independent plane-wave calculations on model-ranked candidates; it located the barrier in per-element reference conventions and confirmed that model ranking concentrates first-principles effort well above the pool base rate. The remaining step is convention matching, not new machinery.

VI. CONCLUSION

Impurity formation-energy prediction with DART within IMP2D is organized by three physical regularities. First, transfer is hierarchical and the hierarchy tracks structural novelty: error grows only mildly from random interpolation (0.408 eV) through withheld host–impurity pairs (0.442 eV) and the predefined chemistry block (0.483 eV), but roughly doubles for a withheld impurity (0.845 eV) or host (0.891 eV), and the host–impurity asymmetry follows the structural novelty of the held-out host after controlling for training support. Second, the two incorporation classes fail for different reasons: interstitial error concentrates in roomy, loosely constrained cages while adsorbate error tracks chemical mismatch, and the measured descriptors attenuate the class gap by roughly a quarter without exhausting it. Third, the formation energy's non-extensive, defect-local character constrains model design: additive pooling degrades withheld-host error severely, a mean readout alone recovers most of that gap, and the defect-centered readout shows no size-dependent residual drift.

These conclusions rest on a representation whose permutation invariance and exact minimum-image distances passed an independent acceptance audit, after a diagnostic showed the superseded implementation could move single predictions by up to 8.8 eV under atom relabeling while pooled errors moved imperceptibly. For existing DFT-relaxed candidates, out-of-fold predictions selected the lower-energy incorporation class well above its class-prior reference. Because the geometries and reference energies arise from the same completed DFT workflow, these results support retrospective energy estimation and reranking within the neutral, retained IMP2D domain rather than a reduction in relaxation cost. Testing independently generated, convention-matched structures and calibrating uncertainty under explicit chemical shift remain necessary before prospective use beyond that domain. The methodology, by contrast, travels as it stands: audit the representation before believing the benchmark, group the evaluation by the chemistry it must survive, and preregister the error analysis — defaults we suggest for machine-learned defect energetics generally.

ACKNOWLEDGMENTS

This work was supported by the National College Students Innovation and Entrepreneurship Training Program (Key Support Area Project). Computations were performed on an NVIDIA L40S cluster at Wuhan University. OpenAI Codex (GPT-5, accessed August 2026) assisted with implementation and debugging of analysis and verification scripts, manuscript restructuring, language revision, and consistency checks. OpenAI GPT Image 2 was used to edit the two schematic figures identified in their captions. The authors directed these uses, reviewed the AI-assisted code, analyses, text, and schematics, and verified the manuscript against the archived results, source code, and cited literature. No AI tool generated or modified the IMP2D reference labels or any quantitative result figure, and the authors take responsibility for the final manuscript.

DATA AVAILABILITY

The source data are available as version 2 of the Interstitial and Adsorbate Structure Database [29]; their construction is described in Ref. [3]. The sample audit, split definitions, resolved configurations, analysis code, and machine-readable results used in this study are available in the project repository [30].

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